

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Zoology

Questions: 29

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. What is Retrogressive Metamorphosis ? Discuss briefly with reference to life history of Ascidians 10 Compare the functional anatomy of the brain in Calotes, Columba and Ottyctolagus. 15 Write an essay on dentition in mammals 15

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2. [UNKNOWN] paper-i

Q2. (a) Schematically present the process of conjugation in Paramoecium sp. Comment on its biological significance. 10+5=15 Define metagenesis. Elaborate the process of metagenesis in Obelia with suitable diagram. 3+12=15 Sphenodon is considered to be a living fossil — Justify. 10

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3. [UNKNOWN] paper-i

Q3. What is Torsion ? Describe the significance of torsion and detorsion in Gastropods. 10 Write an essay on modes of reproduction in Protozoa. 15 Describe the larval forms in Echinodermata. Add a note on their evolutionary significance. 15

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4. [UNKNOWN] paper-i

Q3. (a) Explain the neuro-endocrine regulation of insect metamorphosis with suitable illustrations. Comment on the role of JH (Juvenile Hormone) in insect metamorphosis. 10+5=15 Give an account of accessory respiratory structures in fishes with a note on their biological significance. 15 What is emergency hormone ? How is the secretion of this hormone being regulated? 4+6=10

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5. [UNKNOWN] paper-i

Q4. Describe the biting and chewing, siphoning and piercing and sucking mouth parts in insects with suitable examples. 15 Describe in detail, the life cycle and pathogenicity of Leishmania donovani. 15 Write an account of the canal system in sponges. 10 FSI-P-ZOLY 2 ===== PAGE 3 ===== SECTION B

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6. [UNKNOWN] paper-i

Q4. (a) Give a comparative account of aortic arches in different Vertebrates with suitable illustrations. 15 Peripatus is a living connecting link between Annelida and Arthropoda. Explain. 15 Give the scientific name of one amphibious mollusc. How does it respire in aquatic medium? 2+8=10 FSI-P-ZOLY 2 ===== PAGE 3 ===== SECTION B

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7. [UNKNOWN] paper-i

Q5. Differentiate between the following: 8x5=40 Bioremediation and Biomagnification 8 Correlation and Regression 8 Biological Clock and Biological Rhythm 8 SEM and TEM 8 Infectious and Communicable Diseases 8

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8. [UNKNOWN] paper-i

Q5. (a) Differentiate between culture and capture fisheries. 8 Comment on the prospects of Sericulture in India. 8 Write a brief note on Climax Community. 8 Distinguish between hormone and pheromone with a note on alarm hormone. 8 Write a brief note on the principle and applications of a spectrophotometer. 8

9. [UNKNOWN] paper-i

Q6. "A population is a self-regulating system." Justify. 15 What is Learning Behaviour ? Explain the terms Imprinting, Conditional behaviour and Habituation with suitable examples. 15 Write an essay on Pearl culture in India. Is it an investment avenue? Comment. 10

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10. [UNKNOWN] paper-i

Q6. (a) Give an account on the forest types of India. Why do tropical rainforests support a large number of life forms? 8+7=15 What is population growth ? Delineate different factors affecting population growth. Distinguish between "J" shaped and "S" shaped growth curve with examples. 2+8+5=15 Mention the aims and objectives of "Project Tiger". Comment on its limitations. 5+5=10

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11. [UNKNOWN] paper-i

Q7. What is Pollution ? Describe water pollution, its sources and effect on human health. 15 What is the causative agent of Tuberculosis ? Describe the characteristics, pathogenicity and public health significance. 15 What are Natural Resources ? Mention any two and discuss the conservation strategies undertaken. 10

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12. [UNKNOWN] paper-i

Q7. (a) Why do birds migrate ? Give an account of different methods of navigation during bird migration. 6+9=15 Briefly narrate different techniques of courtship display among birds. 15 Give an account on the echolocation in bats. 10

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13. [UNKNOWN] paper-i

Q8. What are Mean, Median and Mode ? Explain the relationship between these three measures of central tendency. 15 What is Scintillation Counting? Explain the principle and its use in biology. 15 What are the possible consequences of ozone layer depletion? 10 FS!-P-ZOLY 3

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14. [UNKNOWN] paper-i

Q8. (a) Mention the nature of damage caused by *Pyrilla* sp. to sugarcane. Describe the lifecycle stages of the said pest with proper diagram Suggest measures to control the pest. 5+6+4=15 Name important malaria vectors of India. Elaborate the life cycle of any malaria vector of India. How can the transmission of malaria be checked? 4+8+3=15 What is null hypothesis ? Elaborate the application of chi square test in biology. 4+6=10 FSI-P-ZOLY 3

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15. [UNKNOWN] paper-ii

1. Write short notes on the following: 8x5=40 Structure and functions of mitochondria. 8 Watson and Crick model of DNA structure. 8 Genetic basis of ABO blood group system. 8 Chemical and radiation mutagenesis. 8 International - code of zoological Nomenclature. 8

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16. [UNKNOWN] paper-ii

2. Write in detail about the following: 20x2=40 Describe the structure and functions of plasma membrane in cell. 20 What is genome mapping? Discuss the main goals and accomplishments of human genome project. 20

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17. [UNKNOWN] paper-ii

Q2. Discuss the application of pedigree analysis in identifying diseases 15 What are lysosomes ? Discuss their role in

intracellular digestion. Give an example of lysosomal storage disorder. 15 Describe the steps involved in gene regulation in eukaryotes. 10

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18. [UNKNOWN] paper-ii

3. Write in detail about the following: 20x2=40 What are different isolating mechanisms? Discuss their role in speciation. 20 What is Hardy-Weinberg law? Discuss how this law explains the mechanism of evolution. 20 fsi-d-zoly ===== PAGE 2 =====

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19. [UNKNOWN] paper-ii

Q3. With a neatly labelled diagram, explain sodium-potassium pump and its role in membrane transport. 10 Explain the mode of speciation in evolution with examples. 15 Explain how protein synthesis is the key to the expression of biological information. 15

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20. [UNKNOWN] paper-ii

4. Answer the following: 10x4=40 What is protein synthesis? Discuss the role of messenger RNA and ribosomes in protein synthesis. 10 Write down the application of cladistics analysis in summarizing phylogenetic relations. 10 What is DNA fingerprinting? Explain the principles and applications of DNA fingerprinting. 10 What are zoogeographical realms? Describe the salient features of Indo-Malayan realm. 10 SECTION 'B'

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21. [UNKNOWN] paper-ii

Q4. Explain how unknown five-toed Condylarthra evolved in Equus. State the evolutionary pattern in each geological era. 10 Discuss the structure and functions of Ribosomes and Golgi bodies with an emphasis on secretion of proteins. 15 Describe the mechanism of DNA replication. 15 FSI-D-ZOLY 2 ===== PAGE 3 ===== SECTION B

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22. [UNKNOWN] paper-ii

5. Write short notes on the following: 8x5=40 Glycolytic pathway 8 Properties of water soluble vitamins. 8 Structure and functions of immunoglobulin G. 8 Structure and functions of a mammalian nephron. 8 Excitatory and inhibitory neurotransmitters. 8

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23. [UNKNOWN] paper-ii

Q5. (a) Explain enzyme kinetics with suitable examples. 8 Explain how the biological energy transformations obey the laws of thermodynamics. 8 Explain the extrinsic pathway for initiation of blood clotting. 8 Describe the respiratory regulation of acid-base balance in body fluids. 8 What is neoteny ? Discuss the phenomenon in Amphibia giving suitable example. 8

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24. [UNKNOWN] paper-ii

6. Write in detail about the following: 20x2=40 What is blood coagulation? Describe the intrinsic pathway of blood coagulation. 20 6.(b) Describe the different types of human muscles and their properties. 20

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25. [UNKNOWN] paper-ii

Q6. (a) Draw the structure of cAMP. Explain how cAMP acts as second messenger for fl-adrenergic receptor system. 15 Describe the molecular mechanism of skeletal muscle contraction. 10 Define cloning. Explain somatic cell nuclear transfer method with suitable examples 15

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26. [UNKNOWN] paper-ii

7. Comment on the following: 10x4=40 Monosaccharides and disaccharides. FO Saturated and unsaturated fatty acids. 10 Steroid and peptide hormones. 10 Choriovitelline and chorioallantoic placentas. 10

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27. [UNKNOWN] paper-ii

Q7. (a) Discuss the transport of oxygen in the blood. Give an account of various factors that shift oxygen-hemoglobin dissociation curve. 15 Explain enzyme specificity with suitable examples. How is enzyme activity controlled? 15 Describe the process of metamorphosis in frogs and explain the role of thyroxine. 10

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28. [UNKNOWN] paper-ii

8. Answer the following: 10x4=40 Describe the fate map of gastrula with reference to chick. 10 Discuss the basic principles of teratogenesis. 10 Describe in vitro fertilization and embryo transfer techniques. 10 Elaborate the Baer's laws of embryology. 10 fsi-d-zoly 2

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29. [UNKNOWN] paper-ii

Q8. (a) What is embryonic induction ? Describe the mechanism of induction giving examples. 15 Enlist the composition of bile. Discuss the role of bile salts in fat digestion and absorption. 15 What is immunity? Giving suitable examples explain humoral and cell mediated immunity. 10 FSI-D-ZOLY 3

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